

Sofya Lukashevich

 email  GitHub  LinkedIn

Education

KU Leuven Master of Bioinformatics	Sep 2023 – Present Leuven
National Research University Higher School of Economics (HSE)  Ranked 1 st among BSc students in Cellular and Molecular Biotechnology.	Sep 2019 – Jun 2023 Moscow
Uppsala University (Exchange Program) Received a competitive HSE endowment scholarship	Jan 2022 – Jun 2022 Uppsala

Research Experience

Thesis Student, Laboratory of Bioengineering and Morphogenesis <i>KU Leuven</i>	Sep 2024 – Present Leuven
- Developed a pipeline for spatial transcriptomics on a novel chip. - Analyzed neural tube patterning. Conference:  SCSO25 (2025) — <i>Exploring spatially encoded cell fate during human neural tube morphogenesis using Nova-ST</i>. Ramis-Bravo, E.; Mattolini, L.; Lukashevich, S.; et al. [abstract].	
Laboratories of Genetic Engineering and Cell Line Development <i>International Biotechnology Center (IBC) “Generium”</i>	Jun 2022 – Aug 2022 Vladimir oblast’
Engaged in full-cycle R&D of two novel orphan drug candidates for rare diseases.	
Research Intern, International Laboratory of Microphysiological Systems <i>HSE University</i>	Sep 2021 – Oct 2022 Moscow
- Developed a differential equation model for SARS-CoV-2 infection dynamics, improving virus-host interaction insights used in research studies. - Conducted single-cell RNA-seq analyses to examine transcriptional changes during infection.	
Research Assistant, Laboratory of Physiology and Biochemistry of Microorganisms <i>Lomonosov Moscow State University</i>	Sep 2020 – Aug 2021 Moscow
- Assessed 7 hemolytic characteristics of <i>Aspergillus terreus</i> extracellular proteases for biotechnological applications. - Presented project results at  ISCOMS 2021 (Groningen, Netherlands).	

Scientific Publications

-  **CAGEcleaner: Reducing Genomic Redundancy in Gene Cluster Mining.**
Lucas De Vrieze, Miguel Biltjes, Sofya Lukashevich, Kodai Tsurumi, Joleen Masschelein [*bioRxiv Preprint*]
- * Developed a computational tool to detect and eliminate redundant genomic sequences in gene cluster analysis.
-  **Development of a Novel Mathematical Model Explaining SARS-CoV-2 Infection Dynamics in Caco-2 Cells.**
Staroverov, V., Nersisyan, S., Galatenko, A., Lukashevich, S., et al. (2023). *PeerJ*
- * Built a mathematical model to describe SARS-CoV-2 replication dynamics in host cells.
 - * Integrated single-cell RNA-seq data to predict viral progression and host response.

Leadership Experience

Team Captain, Holy Hack 2025 <i>KU Leuven</i>	Mar 2025 Leuven, Belgium
Led a 5-member team in a 24-hour hackathon, delivering a prototype that ranked in the top of the submissions	
Team Captain, Datathon 2025: LLM Deployment on Climate Change Data <i>KU Leuven</i>	Feb 2025 Leuven, Belgium
Coordinated a three-member team to develop a Large Language Model-based chatbot  Chatbot for streamlined climate data analysis and  an Interactive Dashboard for multi-source data integration.	

- Directed bioinformatics efforts that led to a  **Gold Medal win and 2 special prizes (Best Measurement, Inclusivity)** among 438 competing teams.

Awards

Maastricht University & Holland High Potential Scholarship Nominee 2023
Maastricht University

- Selected among the top 2% of applicants for the prestigious Holland High Potential Scholarship.

 **Student Expedition Contest Winner** Jul 2022 – Aug 2022
Kamchatka Peninsula, Russia

- Secured a €30,000 grant for an 11-student expedition to the Kamchatka Peninsula.
- Oversaw expedition logistics, including route planning, transportation, and meal organization.

‘Research Initiative’ Team Project Competition Winner Apr 2021 – May 2021
Krasnodar region, Russia

- Contributed to a seven-member team that won an award for an innovative project on fungal enzymes for wound therapy, funded by HSE University.

Workshops & Professional Development

- **Workshop: Hands-on Introduction to Targeted Spatial Transcriptomics Data Analysis** Dec 2024, VIB (Leuven)
- **Symposium: Single Cell Day - 3rd edition** Oct 2024, LISCO (Leuven)
- **Workshop: Using MOFA for Integration of Omics Data** Sep 2024, VIB (Leuven)
- **Workshop: Sustainability – Johnson & Johnson** Mar 2023, BEST Soft Skills Track (Leuven)
- **Specialization in Genomic Data Science** Coursera (Online)
- **2nd & 3rd Summer Schools on Machine Learning in Bioinformatics** Aug 2021 and 2022, HSE (Moscow)

Skills

Problem-Solving & Analytics

- Data-driven decision-making and AI-powered analysis.
- Strategic problem-solving in interdisciplinary teams.

Languages & Communication

- English (Fluent), Hebrew (Fluent), Russian (Native).
- Dutch (Beginner), Italian (Beginner), French (Beginner).
- Public speaking & science communication experience (conferences, grant pitches).

Technical Skills

- **Programming:** Python, R, SQL, Bash, Java, MATLAB Simulink.
- **Version Control:** Git, GitHub.
- Adobe Photoshop, Illustrator, Figma.

Interests

- **Athletics:** Gymnastics, calisthenics, skiing.
- **Photography:** Film development, cyanotype printing.